WEEK 2 · LESSON 12 OF 24 · HANDOUT

Week 2 Project: Explore a Dataset

Study time: **about 75 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=r-uOLxNrNk8 — open the lesson online to watch inside the course.

Your goal in this lesson: Carry out a complete exploratory data analysis (EDA): load, inspect, clean, summarise and chart a dataset, then report three findings in plain language.

The dataset: moto taxi trips in Kigali

You are an analyst for a (fictional) moto-taxi company. It wants to understand its trips before building a fare-prediction model in Week 4. The data is **simulated** for practice — realistic, but invented. Run this cell:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n = 300
distance = rng.uniform(1, 15, n).round(1)
rain = rng.random(n) < 0.25
hour = rng.integers(6, 23, n)
fare = (400 + 230 * distance + np.where(rain, 300, 0) + rng.normal(0, 150, n)).round(-1)
minutes = (distance * 2.8 + rng.normal(0, 3, n) + np.where(rain, 4, 0)).clip(3).round()
trips = pd.DataFrame({
    "distance_km": distance,
    "minutes": minutes,
    "fare_rwf": fare,
    "hour": hour,
    "raining": rain,
    "district": rng.choice(["Gasabo", "Kicukiro", "Nyarugenge"], size=n),
})
# a few realistic data problems
trips.loc[[5, 77, 140], "fare_rwf"] = np.nan
trips.loc[200, "distance_km"] = 150.0
trips = pd.concat([trips, trips.iloc[[10, 11]]], ignore_index=True)
```

Part 1 — Inspect (Lesson 8)

Run `head()`, `shape`, `info()` and `describe()`. Write down: how many trips, which columns, which types, and anything that looks strange.

Part 2 — Clean (Lesson 9)

Find and handle: - the duplicate rows, - the missing fares, - the impossible distance.

Write a short cleaning log (what you changed and why).

Part 3 — Summarise (Lesson 10)

Answer with code: 1. What is the average fare and the average distance? 2. How much more is the average fare when it is raining? 3. Which district has the longest average trip? 4. Which hours of the day are busiest?

Part 4 — Chart (Lesson 11)

Make at least three clear charts, each with a title and labelled axes: - a histogram of fares, - a scatter plot of distance vs fare (colour rainy trips differently: `c=trips["raining"]`), - a bar chart of trips per hour.

Part 5 — Report

Write a half-page report for the company's manager:

- **Three findings**, each with a number (for example, *"Rainy trips cost about 300 RWF more on average."*).
- **One chart** that best supports your findings.
- **One question** the data cannot answer yet, and what data would be needed.

Example code to get started

```
clean = trips.drop_duplicates()
clean = clean[clean["distance_km"] <= 40]
clean = clean.dropna(subset=["fare_rwf"])
print(len(trips), "->", len(clean))
print(clean.groupby("raining")["fare_rwf"].mean().round())
```

Checklist

- Every step has its own cell with a short text note above it.
- The cleaning log lists every change.
- Charts have titles, axis labels and units.
- Findings use simple English and real numbers from your analysis.

Remember this: Exploratory data analysis is a loop of looking, cleaning, summarising and charting. It always comes before modelling — it tells you whether the data can answer the question at all.

Name: _____ Date: _____

Activity

Task 1

Complete Parts 1–5 in one Colab notebook. In Week 4 you will build a model that predicts `fare_rwf` from `distance_km` using this same data, so keep your notebook.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is exploratory data analysis (EDA)?
   - A. Training a neural network
   - B. Looking at, cleaning, summarising and charting data to understand it before modelling
   - C. Publishing a website
   - D. Deleting old data
2. A trip has distance_km = 150 inside Kigali. What is it most likely to be?
   - A. A normal trip
   - B. A missing value
   - C. A long-distance record to celebrate
   - D. A data error (impossible value) to fix or remove
3. Which chart best shows whether fare rises with distance?
   - A. Scatter plot
   - B. Table of districts
   - C. Pie chart
   - D. Histogram of hours
4. Which code compares average fares for rainy and dry trips?
   - A. clean.sort_values("raining")
   - B. clean["fare_rwf"].sum()
   - C. clean.groupby("raining")["fare_rwf"].mean()
   - D. clean["raining"].plot()
5. What makes a good finding in the report?
   - A. A long list of all columns
   - B. A vague feeling
   - C. Only a chart without words
   - D. A clear statement backed by a number from the analysis
6. Why is EDA done before building a model?
   - A. To make the notebook longer
   - B. Models cannot read clean data
   - C. It is not needed
   - D. It shows whether the data is clean and able to answer the question

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